

DSU Client Project

Tech Investment Bank

Online M&A Advisory Platform



Our Team



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Background

- Our client is a specialized mergers and acquisitions business to help other companies sell, with clients across many industries from hospitality to tech
- They provide a system for companies to find buyers in fields such as Private equity to vertical partners.
- Private companies submit their information through a questionnaire → experts create a valuation for the company

Project Goals

Data Extraction

Obtain private company data from questionnaire (EBITDA, revenue, industry, location, etc)

Prompting

Develop a script to automate queries to Perplexity AI, integrating private company data

Comparisons

Identify similar public companies in order to find a comparable valuation

Historical Dashboarding

Gather public company information and create financial evaluation + dashboard

Prompting w/ Perplexity AI

- Method: leverage Perplexity API calls (instead of GUI) for smoother data pipelining
- Input: JSON - info of private company
- Output: JSON objects of 5 most similar public companies, in format we prompt
- Reflection
 - Sonar-pro vs Sonar-reasoning-pro

```
prompt = f"""
Given a private company with the following characteristics:
{private_company}

Please return a JSON object containing 5 most relevant public company co
{{
  "comparable_companies": [
    {{
      "ticker": "[STOCK TICKER]",
      "company_name": "[FULL LEGAL NAME]",
      "selection_rationale": "[1-2 sentences on why this company i
    }}
  ]
}}

Prioritize companies that:
1. Share similar business models and revenue streams
2. Operate in similar geographic markets
3. Have similar growth profiles
4. Are pure-play companies in the same industry
```

Sonar-reasoning

“Premier reasoning offering powered by DeepSeek R1 with Chain of Thought”

"<think>Okay, I need to find five public company comparables for a private SaaS company that provides AI-driven data analytics solutions for financial institutions. The company has \$500M in revenue, 20% EBITDA margin, and a 15% growth rate. The primary NAICS code is 511210, which is Software Publishers, and they focus on North America. First, I should look for public companies in the same industry (NAICS 511210) that offer SaaS solutions, ..."

Sonar-pro

```
Ticker: SNOW
Company Name: Snowflake Inc.
Rationale: Snowflake provides a cloud-based data platform with a strong emphasis on financial services clients. It is highly comparable to the subject company, though Snowflake may be larger in scale.

Ticker: MDB
Company Name: MongoDB, Inc.
Rationale: MongoDB offers a database platform used by many financial institutions for analytics. Its SaaS model, and focus on financial services, make it highly comparable, despite being more focused on database management than analytics.

Ticker: PLTR
Company Name: Palantir Technologies Inc.
Rationale: Palantir provides AI-powered data analytics platforms, with a significant presence in the financial services industry. Its focus on data analytics and AI makes it highly comparable, though it may have a broader industry focus than the subject company.
```

Public Company Extraction

Private Comp Info

Extracted information about private companies (revenue, margin, demographics) from a JSON file of questionnaires.

Revenue: \$---
Ebitda_margin: %--
Growth Rate: %--
Description: "Hotel retreat center in Peru ..."
Industry: ----
Where it's based: ---

Perplexity AI

Built and tested prompts to feed into Perplexity AI with private company as input.



Results

For each company, our script output the tickers of the 5 most similar public companies for each private company.

Example Output

['MAR', 'H', 'ABNB', 'PLYA', 'STAY']

Ticker Metrics Extraction

These are the metrics to be inputted into the client's valuation software

EBITDA

Earnings before interest,
taxes, depreciation, and
amortization)

Enterprise Value

Total market value of company
→ Equity market capitalization
+ debt

Total Revenue

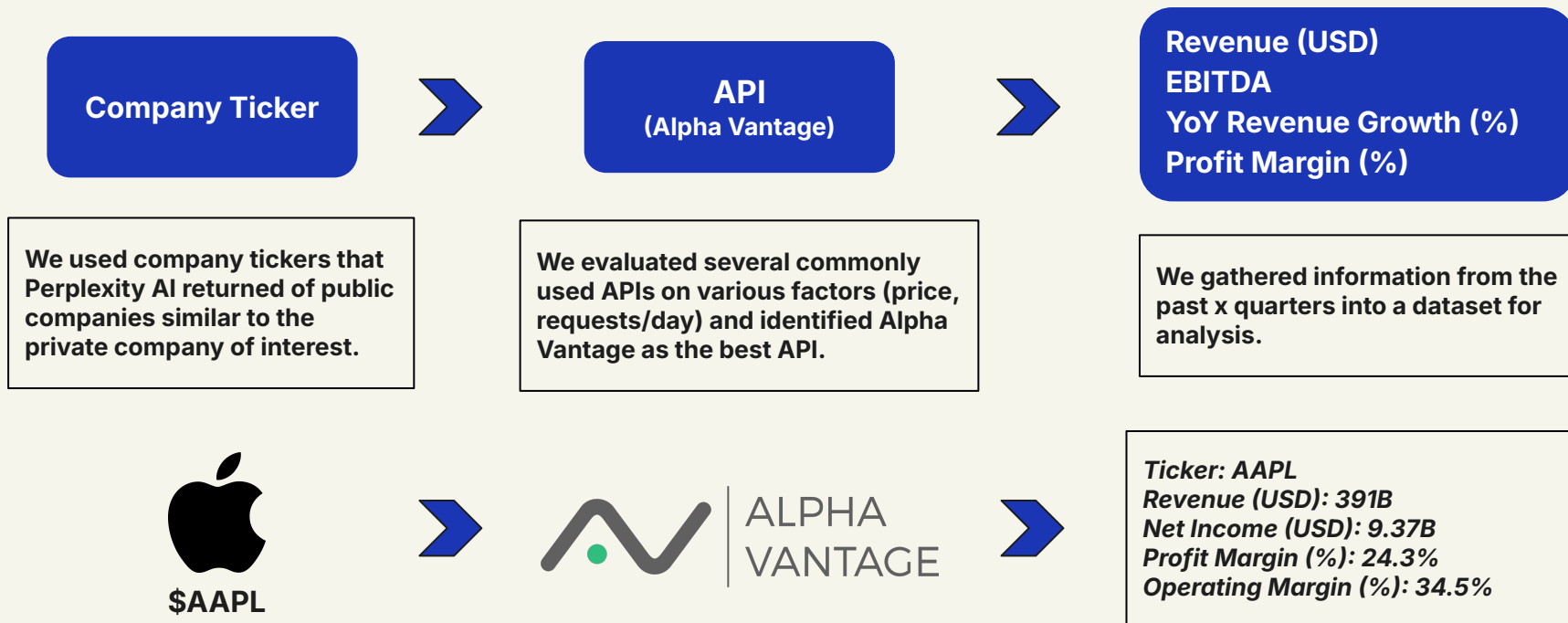
Revenue generated per year

Year-over-year revenue growth %

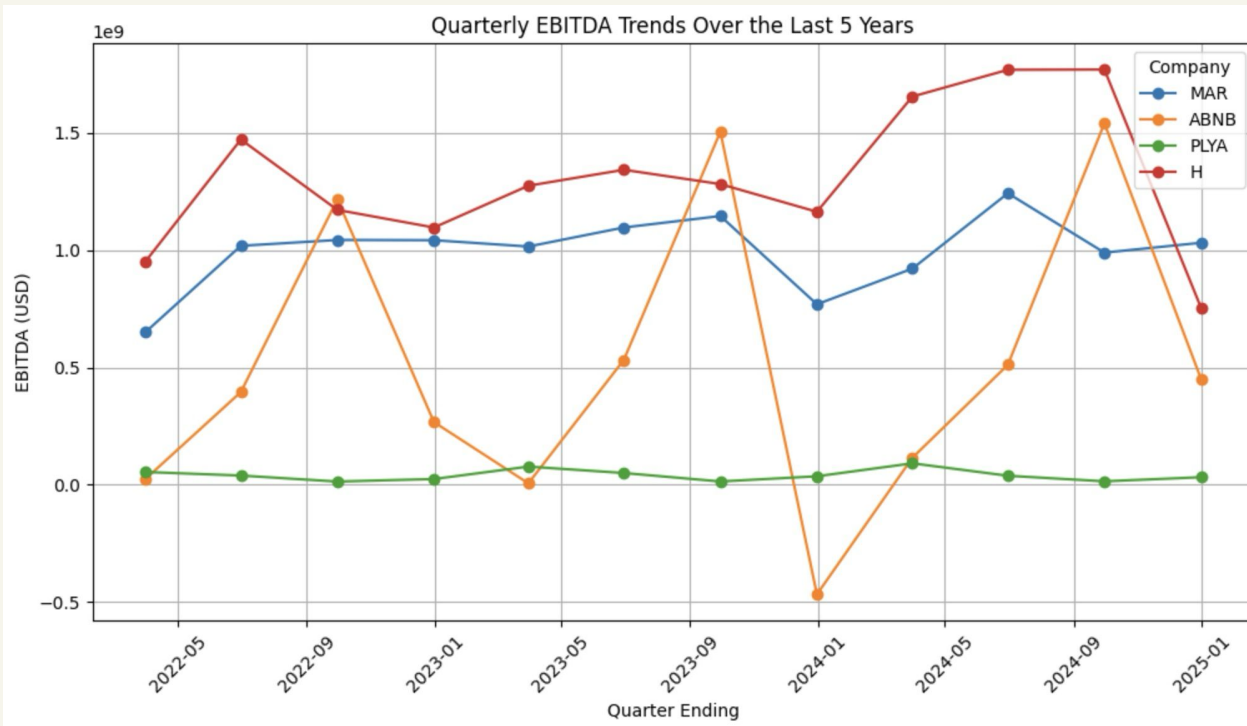
Profit Margin %

Net income over revenue

API Financial Extraction



Dashboarding



Using the data from our financial extraction calls, we create a dataset of 5 years of historical data

We then create a dashboard of historical EBITDA trends for the public comparisons

Thank you!

Any questions?